Module 13: How Populations Evolve - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Microevolution is defined as:
    - A. the origin of entirely new kingdoms of life
    - B. a change in allele frequencies in a population over generations
    - C. a change in a single individual's body
    - D. the extinction of every species
    - Answer: B
    - Why: Microevolution is a shift in allele frequencies within a population over time.
2. Genetic drift has its strongest effect in:
    - A. small populations
    - B. very large populations
    - C. populations with no variation
    - D. populations that never reproduce
    - Answer: A
    - Why: Random sampling changes allele frequencies most strongly when the population is small.
3. A few birds colonize a new island and start a population with limited variation. This reduced variation is due to:
    - A. natural selection alone
    - B. gene flow between islands
    - C. mutation pressure
    - D. the founder effect
    - Answer: D
    - Why: The founder effect is genetic drift that occurs when a new population begins from only a few individuals.

4. The Hardy-Weinberg principle is useful mainly because it:

    - A. proves evolution never happens
    - B. measures average body size
    - C. provides a non-evolving baseline to compare real populations against
    - D. edits alleles directly
    - Answer: C
    - Why: Hardy-Weinberg predicts genotype proportions under stated assumptions. A deviation calls for checking mating, sampling, and evolutionary processes; it does not alone prove allele frequencies changed.